

Jaechul Roh

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EDUCATION

University of Massachusetts Amherst

Ph.D. in Computer Science

Advisor: Prof. [Amir Houmansadr](#)

GPA: 4.0/4.0

September 2023 – May 2028 (expected)

Amherst, Massachusetts, USA

Hong Kong University of Science and Technology

B.Eng. in Computer Engineering, School of Engineering

Final Year Thesis Advisor: Prof. [Jun Zhang](#)

*2 years of Compulsory Korean Military Duty

September 2017 – May 2023

Clear Water Bay, Hong Kong

RESEARCH INTERESTS

My research focuses on the **Privacy & Security of AI models and agents**. Under the supervision of Prof. Amir Houmansadr, I study the trustworthiness of multimodal generative models across audio, text, and vision domains, while also collaborating closely with Prof. Eugene Bagdasarian on several projects and publications. Most recently, I completed a Summer Research Internship at Brave Software under the guidance of Dr. Ali Shahin Shamsabadi, where our work centered on advancing privacy in AI web agents.

PUBLICATIONS

Under Review

1. **SPILLage: Agentic Oversharing on the Web**

Jaechul Roh, Eugene Bagdasarian, Hamed Haddadi, Ali Shain Shamsabadi

Under Review

2. **Bob's Confetti: Phonetic Memorization Attacks in Music and Video Generation**

Jaechul Roh, Zachary Novack, Yuefeng Peng, Niloofar Miresghallah, Taylor Berg-Kirkpatrick, Amir Houmansadr

Preprint at arXiv (Under Review)

[\[paper\]](#) [\[demo page\]](#)

3. **OverThink: Slowdown Attacks on Reasoning LLMs**

Abhinav Kumar, **Jaechul Roh**, Ali Naseh, Marezna Karpinska, Mohit Iyyer, Amir Houmansadr, and Eugene Bagdasarian

Preprint at arXiv (Under Review)

[\[paper\]](#) [\[code\]](#)

4. **World-To-Image: Grounding Text-to-Image Generation with Agent-Driven World Knowledge**

Moohyun Son, Jintaek Oh, Sunbin Mun, **Jaechul Roh**, Sehyun Choi

Under Review

5. **Throttling Web Agents Using Reasoning Gates**

Abhinav Kumar, **Jaechul Roh**, Ali Naseh, Amir Houmansadr, Eugene Bagdasarian

Preprint at arXiv (Under Review)

[\[paper\]](#) [\[code\]](#) [\[demo page\]](#)

Conference

1. **Multilingual and Multi-Accent Jailbreaking of Audio LLMs**

Jaechul Roh, Virat Shejwalkar, Amir Houmansadr

COLM 2025

[\[paper\]](#)

2. **Backdooring Bias (B^2) into Stable Diffusion Models**

Ali Naseh, **Jaechul Roh**, Eugene Bagdasarian, Amir Houmansadr

USENIX Security '25

[\[paper\]](#) [\[code\]](#)

3. **OSLO: One-Shot Label-Only Membership Inference Attacks**

Yuefeng Peng, **Jaechul Roh**, Subhransu Maji, Amir Houmansadr

NeurIPS 2024

[\[paper\]](#)

4. **Memory Triggers: Unveiling Memorization in Text-To-Image Generative Models through Word-Level Duplication**
Ali Naseh, **Jaechul Roh**, Amir Houmansadr
The 5th AAAI Workshop on Privacy-Preserving Artificial Intelligence
[\[paper\]](#)
5. **Robust Smart Home Face Recognition under Starving Federated Data**
Jaechul Roh, Yajun Fang
IEEE International Conference on Universal Village (IEEE UV2022)
Oral Presentation
[\[paper\]](#)[\[code\]](#)[\[slides\]](#)[\[video\]](#)
6. **MSDT: Masked Language Model Scoring Defense in Text Domain**
Jaechul Roh, Minhao Cheng, Yajun Fang
IEEE International Conference on Universal Village (IEEE UV2022)
Oral Presentation
[\[paper\]](#)[\[code\]](#)[\[slides\]](#)[\[video\]](#)
7. **Impact of Adversarial Training on the Robustness of Deep Neural Networks**
Jaechul Roh
2022 IEEE 5th International Conference on Information Systems and Computer Aided Education (ICISCAE)
[\[paper\]](#)[\[code\]](#)

Preprints

1. **Chain-of-Code Collapse: Reasoning Failures in LLMs via Adversarial Prompting in Code Generation**
Jaechul Roh, Varun Gandhi, Shivani Anilkumar, Arin Garg
Preprint at arXiv
[\[paper\]](#) [\[code\]](#)
2. **R1dacted: Investigating Local Censorship in DeepSeek's R1 Language Model**
Ali Naseh, Harsh Chaudhari, **Jaechul Roh**, Mingshi Wu, Alina Oprea, Amir Houmansadr
Preprint at arXiv
[\[paper\]](#)
3. **FameBias: Embedding Manipulation Bias Attack in Text-to-Image Models**
Jaechul Roh*, Andrew Yuan*, Jinsong Mao*
Preprint at arXiv (Equal Contribution)*
[\[paper\]](#)
4. **Understanding (Un)Intended Memorization in Text-to-Image Generative Models**
Ali Naseh, **Jaechul Roh**, Amir Houmansadr
Preprint at arXiv
[\[paper\]](#)

Magazine Article

1. **Can You Trust What AI Hears (and Says)?**
Jaechul Roh
XRDS: Crossroads, The ACM Magazine for Students 31 (4), 34-39
[\[article\]](#)

INVITED TALKS & ACADEMIC SERVICES

Google Speech Technologies Group

Paper presentation

July 2025

Google DeepMind

- Presented our "Multilingual and Multi-Accent Jailbreaking of Audio LLMs" paper to the NFM Reading Group led by the Speech Technologies Group at Google DeepMind.
[\[slides\]](#)

Program Committee Member (Reviewer)

- **Main conference:** ICLR (2025, 2026)
- **Workshop:** COLM XLLM-Workshop (2025)

WORK EXPERIENCE

Brave Software

Research Intern, Supervisor: Ali Shahin Shamsabadi

June 2025 – September 2025

London, United Kingdom (Remote)

- Working on privacy & security of AI agents.

Super Chain AI (Conard International)

NLP Engineer Intern, Supervisor: Cat Yung

June 2021 – August 2021

Kowloon Bay, Hong Kong

- In charge of topic modeling and semantic analysis based on customer reviews and assigning specific semantics to the topics extracted.
- Competitors' analysis through web-scraping customer reviews from other drop-shipping websites.

Military Service at Head Quarter of 12th Infantry Division

Sergeant of Republic of Korea Army

July 2018 – March 2020

Injae, Kang Won Do, Republic of Korea

- Officer Administrative Clerk Specialist
- Squad Leader of the Head Quarter

SKILLS / LANGUAGES

Programming Language: Python, C++

Languages: Korean (Native), English (Native), Chinese (Fluent)